Methods of Proof

Examples and Counterexamples

In this section, we discuss when it is admissible (and appropriate) to (dis)prove a statement by providing a single (counter)example. In each case, you should not provide a general rule which such an example would follow; you should provide a **specific element of the domain** which proves/disproves the given statement.

You **must** clearly explain why your given example works. It is not enough to simply give "an answer", you must also justify that your example satisfies the given conditions.

Proof by Example

Let $P(x)$ be a (possibly very complicated) predicate in the domain U. To prove a statement of the form $(\exists x)P(x)$, you must find a **specific value** $a \in U$ such that $P(a)$ is true.

Example: Prove that there is an even integer which is divisible by 3.

Proof: Let $n = 12$. Then $n = 2 \cdot 6$ so that by definition, n is an even integer. Furthermore, $n = 3 \cdot 4$ so that $3|n$.

Things to observe:

- The proof does not just state $n = 12$. We have to explain *why* $n = 12$ is an example of the claim.
- The proof does not just state that any integer which is a multiple of 2 and a multiple of 3 will work. We provide a *specific* value $n = 12$, and then demonstrate that n has both of the desired properties.

Example: Prove that there is an integer greater than every negative integer.

Proof: Let $n = 0$, and suppose that $x \in \mathbb{Z}$ is negative. Note that $n \in \mathbb{Z}$. Furthermore, since x is negative, by definition $x < 0 = n$ so that $x < n$. Therefore, $n = 0$ is greater than every negative integer.

Things to observe:

- The proof does not just state $n = 0$. We have to explain *why* $n = 0$ is an example of the claim.
- The proof does not just state that any integer which is greater than or equal to 0 will work. We provide a *specific* value $n = 0$, and then demonstrate that n has the desired properties.
- If we write the original claim in predicate logic, this becomes something like $(\exists n)(\forall x)(N(x) \implies x < n)$ where $N(x)$ is the predicate "x is negative". This means that our job is to find a specific example n for which the **whole statement** $(\forall x)(N(x) \implies x < n)$ is true. How do you prove a "$\forall$" statement? (See later section.)

Disproof by Counterexample

Let $P(x)$ be a (possibly very complicated) predicate in the domain U. To **dis**prove a statement of the form $(\forall x)P(x)$, you must prove the negation. That is, to disprove a "$\forall$" statement, you must prove $(\exists x)\neg P(x)$. This means you are proving an "$\exists$" statement, and so all the same notes above apply.

"Universal" Proofs

In this section, we discuss how to prove a statement of the form $(\forall x)P(x)$. Here are some examples of when you should think to use such proofs:

- To prove every integer divisible by 4 is even, this takes the form $(\forall n)(4|n \implies 2|n)$.
- To prove that $A \subseteq B$ for sets A and B, this takes the form $(\forall a)(a \in A \implies a \in B)$.
- To prove that a relation R on a set A is reflexive, this takes the form $(\forall a \in A)(aRa)$.
- To prove that a relation R on a set A is symmetric, this takes the form $(\forall a, b \in A)(aRb \implies bRa)$.
- To prove that a relation R on a set A is transitive, this takes the form $(\forall a, b, c \in A)((aRb \wedge bRc) \implies aRc)$.

- To prove that a function $f : X \to Y$ is an injective function, this takes the form $(\forall a, b \in X)(f(a) = f(b) \implies a = b)$.
- To prove that a function $f : X \to Y$ is a surjective function, this takes the form $(\forall y \in Y)(\exists x \in X)(f(x) = y)$.

Note that proving a relation R is reflexive, and that surjectivity of a function are the only statements here **not** of the form $(\forall x)(P(x) \implies Q(x))$.

Let $P(x)$ be a (possibly very complicated) predicate in the domain U. To prove a statement of the form $(\forall x)P(x)$, you must show that $P(x)$ is true **for an arbitrary** $x \in U$. That is, you should start the proof of such a claim with the statement "Let $x \in U$ be given." (Here, U is, of course, replaced by the domain presented in the problem.) You must then show that the (possibly very complicated) predicate $P(x)$ is true for that particular arbitrary x you are working with.

Here, we give examples of proofs that are **not** $\implies$ statements.

Example: Let R be the relation on $\mathbb{Z}$ defined by aRb if and only if $a^2 = b^2$. Prove that R is reflexive.

Proof: Let $a \in \mathbb{Z}$. Then $a = a$ so that $a^2 = a^2$. Therefore, aRa. Hence, R is reflexive.

Things to observe:

- The proof starts by introducing a as an integer (an element of our domain). Since we are **not** proving an $\implies$ statement, we make no other assumptions about a.
- The proof does not state that aRa until this has been shown based on the given assumptions.
- Every sentence is justified clearly by something preceding it.

Direct Proofs

To prove $(\forall x)(P(x) \implies Q(x))$ using a direct proof, start by introducing your variable x as an element of the domain U, and assume that the premise $(P(x))$ is true. Show that $Q(x)$ must follow. (*Note:* If you start your proof by either saying "Let $x \in U$ and suppose $P(x)$ implies $Q(x)$" or "Let $x \in U$ and suppose $P(x)$ and $Q(x)$", then you are assuming what you are trying to prove. This is a significant error!)

Example: Prove that every integer divisible by 4 is even.

Proof: Let $n \in \mathbb{Z}$ and suppose $4|n$. Then $n = 4k$ for some $k \in \mathbb{Z}$. Then $n = 2(2k)$, and since $2k \in \mathbb{Z}$ by closure under multiplication, by definition n is even. Therefore, if $4|n$, then n is even.

Things to observe:

- The proof starts by introducing n as an integer (an element of our domain) and assuming that $4|n$ (the premise of the $\implies$ statement)
- The proof does not state that n is even until this has been shown based on the given assumptions.
- Every sentence is justified clearly by something preceding it.

Example: Let R be the relation on $\mathbb{Z}$ defined by aRb if and only if $a^2 = b^2$ Prove that R is transitive.

Proof: Let $a, b, c \in \mathbb{Z}$. Suppose that aRb and bRc. By definition, $a^2 = b^2$ and $b^2 = c^2$. By substitution, $a^2 = c^2$, and by definition of R, we have aRc.

Things to observe:

- To prove that R is transitive, we must prove $(\forall a, b, c \in \mathbb{Z})((aRb \wedge bRc) \implies aRc)$.
- The proof starts by introducing $a, b, c \in \mathbb{Z}$ and assuming that aRb and bRc (i.e., the premise of the $\implies$ statement).
- The proof ends by showing that aRc, and therefore that R is transitive.
- The proof does not state that aRc until that has been shown.
- Every sentence is justified clearly by something preceding it.

Indirect Proofs

There are two flavors of indirect proofs

Proof by Contraposition To prove $(\forall x)(P(x) \implies Q(x))$ we can also prove the contrapositive: that $(\forall x)(\neg Q(x) \implies \neg P(x))$. This is then just a proof of the same form as seen in the section "Direct Proofs", but where we must tell the reader we are doing a proof by contraposition.

Example: Prove by contraposition that if n is odd, then n is not divisible by 4.

Proof: Let $n \in \mathbb{Z}$ and suppose for contraposition that $4|n$. Then $n = 2(2k)$, and since $2k \in \mathbb{Z}$ by closure under multiplication, n is even by definition. Since n is even, n is not odd. Therefore, by contraposition, if n is odd then $4 \not| n$.

Things to observe:

- The proof begins by telling the reader we are doing a proof by contraposition.
- The proof ends by telling the reader what the original statement we wanted to prove was, and that we proved it using contraposition.
- The proof starts by introducing n as an integer (an element of our domain) and assuming that $4|n$ (the premise of the **contrapositive** of the original $\implies$ statement).
- The proof does not state that n is even until this has been shown based on the given assumptions.
- Every sentence is justified clearly by something preceding it.

Proof by Contradiction To do a proof by contradiction, you start by assuming that the entire statement you have been given is false. That is, you assume **to the contrary** that the negation is true. Your goal is to show that this leads to two incompatible facts (called a contradition). This contradiction should likely take the form of saying that p is true and p is not true.

Proofs by contradiction likely require you to simplify the negation of a statement before you start your work, so that you know what the "negation" that you are assuming actually is.

Example: Prove that if n^2 is even then n is even.

Proof: Suppose to the contrary that there is some $n \in ZZ$ such that n^2 is even but that n is not even. By our axioms, since n is not even, n must be odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. By substitution, $n^2 = (2k+1)^2 = 4k^2 + 2k + 1 = 2(2k^2 + k) + 1$. By closure under addition and multiplication, $2k^2 + k \in \mathbb{Z}$, and so by definition n^2 is odd. By our axioms, an integer is either even or odd, but not both, but we assumed that n^2 is even, and have shown that n^2 is odd. This is a contradiction. Therefore, if n^2 is even, so is n.

Things to observe:

- The original statement is of the form $(\forall n)(E(n^2) \implies E(n))$ where $E(n)$ means "n is even". The *negation* of this statement is $(\exists n)(E(n^2) \wedge \neg E(n))$. (**I strongly encourage you to prove this is in fact a simplification of the negation using a proof sequence**.)
- The proof begins by telling the reader we are doing a proof by contraposition, and states that we are assuming that the negation of the original statement is true. (That is, we state the negation that is provided in the previous bullet point.)
- The proof ends by telling the reader that we arrive at a contradiction (and clearly states what that contradiction is).
- The proof does not state that we have arrived at a contradiction until that contradiction has been clearly demonstrated.
- After examining the contradiction, we explain what the appropriate conclusion is (i.e., that the negation of the statement was false, and therefore the original statement must be true).
- Every sentence is justified clearly by something preceding it.